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Assignment (Software)

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EEE 321L
Power System I Laboratory
Assignment/Project (Software)
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Problem Statement

The Single Line Diagram (SLD) of the system is shown in *Figure-1*. The data for the components in the system are given in per unit in *Tables - 1, 2 and 3*. The base values are $S_{base}=100$ MVA, $V_{base}=15$ kV at bus 1. The ratings of the devices are:

Generator G1: 100 MVA, 15kV
Generator G2: 100 MVA, 15kV
Transformer T1: 100 MVA, 15 Δ /345 Y kV
Transformer T2: 100 MVA, 345 Y/15 Δ kV

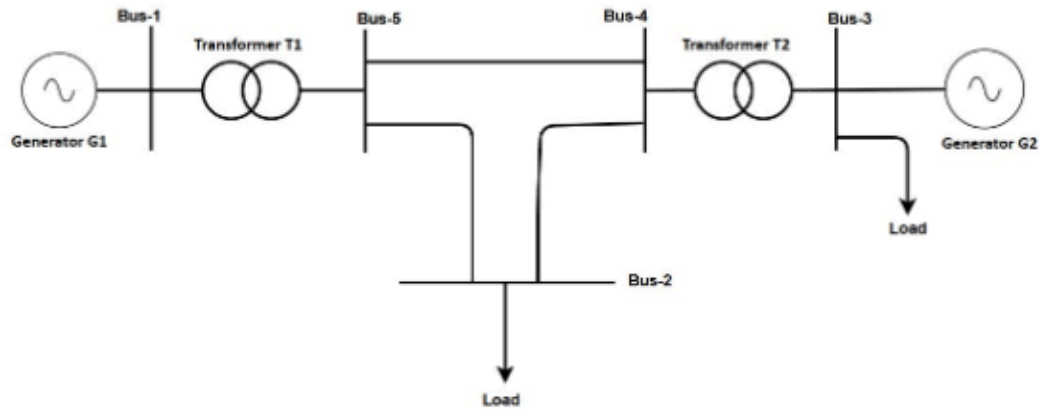


Figure-1: Single Line Diagram of the Power System

Table 1: Bus Data

Bus	Type	V (p.u.)	δ (degrees)	P_G (p.u.)	Q_G (p.u.)	P_L (p.u.)	Q_L (p.u.)	Q_{Gmax} (p.u.)	Q_{Gmin} (p.u.)
1	Swing	1.0	0	—	—	0	0	—	—
2	Load	—	—	0	0	8.0	2.8	—	—
3	Voltage- controlled	1.05	—	5.2	—	0.8	0.4	4.0	-2.4
4	Load	—	—	0	0	0	0	0	—
5	Load	—	—	0	0	0	0	0	—

Table 2: Line Input-Data

Bus-to-bus	R (p.u.)	X (p.u.)	G (p.u.)	B (p.u.)	Maximum MVA (p.u.)
2-4	0.009	0.1	0	1.72	12.0
2-5	0.0045	0.050	0	0.88	12.0
4-5	0.00225	0.025	0	0.44	12.0

Table 3: Transformer Data

Bus-to-bus	R (p.u.)	X (p.u.)	G_c (p.u.)	B_m (p.u.)	Maximum MVA (p.u.)
1-5	0.0015	0.02	0	1.72	6.0
4-3	0.00075	0.01	0	0.88	10.0

Answer the following questions

- (a) Draw the system in *PowerWorld Simulator*.
- (b) Use *PowerWorld* to determine the *Y_{bus}* of this system.
- (c) Perform Load-flow analysis in *PowerWorld* using *Gauss-seidel Method* and from the simulation determine the *State Variables* associated with each bus. Also determine the *Line losses, losses in both the Transformers, Power Consumed by both the Loads*.

Generator Data setup

The screenshot shows the 'Generator Options' dialog box for Generator 1. The 'Bus Number' is 1, 'Bus Name' is 'Bus 1', and 'Area Name' is 1. The 'Status' is 'Closed'. The 'Generator MVA Base' is 100.00. The 'Fuel Type' is 'UN (Unknown)' and 'Unit Type' is empty. The 'Display Information' tab is selected. The 'Power Control' section shows 'MW Setpoint' at 393.900, 'MW Output' at 393.900, 'Min. MW Output' at 0.000, and 'Max. MW Output' at 1000.000. The 'Part. Factor' is 10.00. The 'Available for AGC' checkbox is checked, and 'Enforce MW Limits during automatic control' is also checked. The 'Voltage Control' section shows 'Mvar Output' at -81.804, 'Min Mvars' at -9900.000, and 'Max Mvars' at 9900.000. The 'Available for AVR' checkbox is checked, and 'Use Capability Curve' is unchecked. The 'Regulated Bus Number' is 1, 'SetPoint Voltage' is 1.000000, 'SetPoint Voltage Tol' is 0.000000, and 'Remote RegFactor' is 100.0. The 'Line Drop Compensation' section shows 'Use LDC' set to 'No', 'Xcomp' at 0.000100, and 'Rcomp' at 0.000000. The 'Voltage Droop Control' section is empty. The 'Wind Control Mode' is 'None' and 'Power Factor' is 1.0000. The 'Mvar Capability Curve' table is empty. The 'OK', 'Save', 'Save to Aux', 'Cancel', and 'Help' buttons are at the bottom.

	MW	Min Mvar	Max Mvar
1			
2			
3			
4			
5			

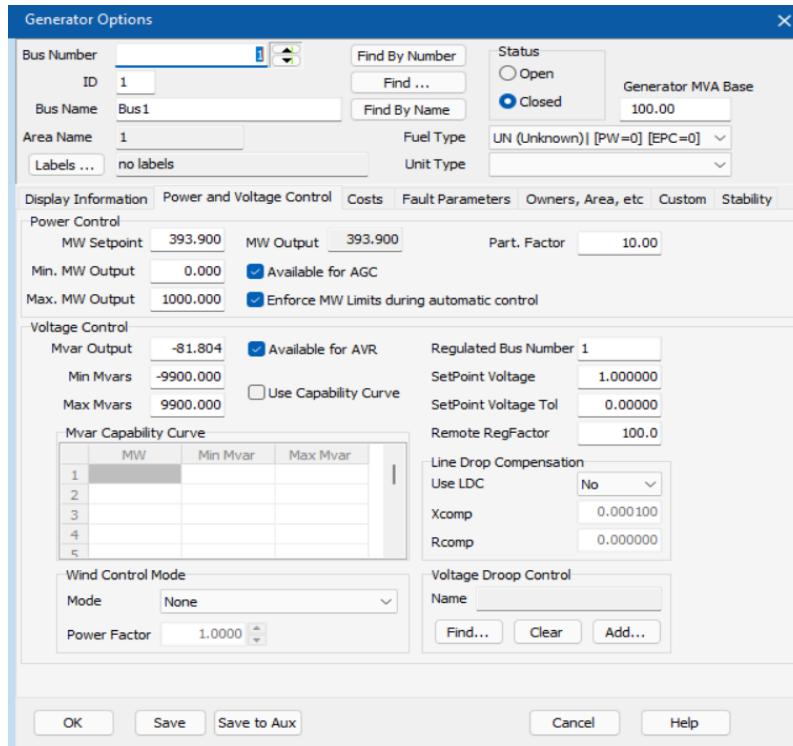
Figure:- Data setup for Generator 1(Connected with BUS 1)

The screenshot shows the 'Generator Options' dialog box for Generator 2. The 'Bus Number' is 3, 'Bus Name' is 'Bus3', and 'Area Name' is 1. The 'Status' is 'Closed'. The 'Generator MVA Base' is 100.00. The 'Fuel Type' is 'UN (Unknown)' and 'Unit Type' is empty. The 'Display Information' tab is selected. The 'Power Control' section shows 'MW Setpoint' at 520.000, 'MW Output' at 520.000, 'Min. MW Output' at 0.000, and 'Max. MW Output' at 1000.000. The 'Part. Factor' is 10.00. The 'Available for AGC' checkbox is checked, and 'Enforce MW Limits during automatic control' is also checked. The 'Voltage Control' section shows 'Mvar Output' at 252.979, 'Min Mvars' at -240.000, and 'Max Mvars' at 400.000. The 'Available for AVR' checkbox is checked, and 'Use Capability Curve' is unchecked. The 'Regulated Bus Number' is 4, 'SetPoint Voltage' is 1.050000, 'SetPoint Voltage Tol' is 0.000000, and 'Remote RegFactor' is 100.0. The 'Line Drop Compensation' section shows 'Use LDC' set to 'No', 'Xcomp' at 0.000100, and 'Rcomp' at 0.000000. The 'Voltage Droop Control' section is empty. The 'Wind Control Mode' is 'None' and 'Power Factor' is 1.0000. The 'Mvar Capability Curve' table is empty. The 'OK', 'Save', 'Save to Aux', 'Cancel', and 'Help' buttons are at the bottom.

	MW	Min Mvar	Max Mvar
1			
2			
3			
4			
5			

Figure:- Data setup for Generator 2(Connected with BUS 3)

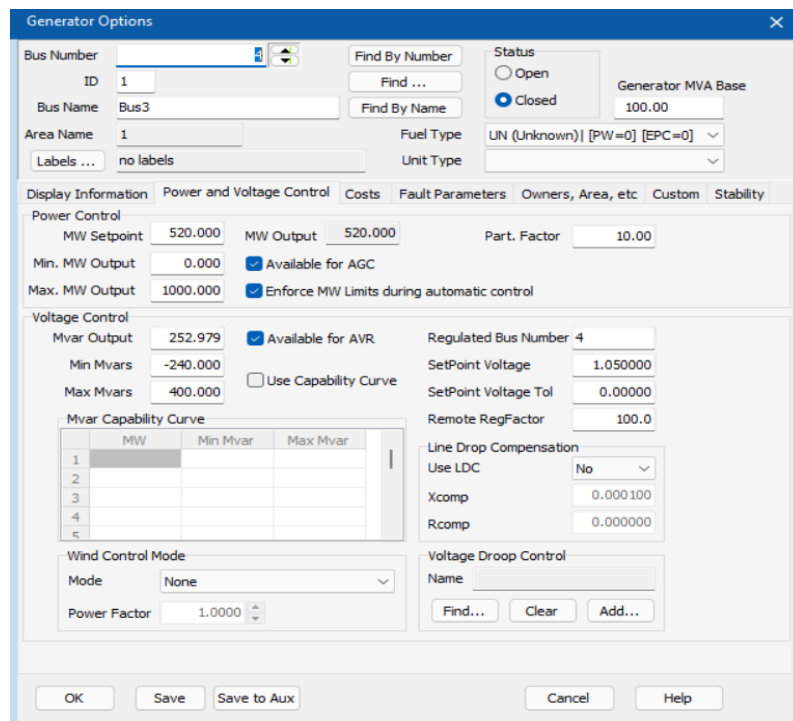
Transformer Data setup



The 'Generator Options' dialog box is shown for Bus 1. The 'Status' is set to 'Closed'. The 'Generator MVA Base' is 100.00. The 'Fuel Type' is 'UN (Unknown)' and the 'Unit Type' is empty. The 'Display Information' tab is selected. The 'Power Control' section shows 'MW Setpoint' at 393.900, 'MW Output' at 393.900, and 'Part. Factor' at 10.00. The 'Voltage Control' section shows 'Mvar Output' at -81.804, 'Min Mvars' at -9900.000, and 'Max Mvars' at 9900.000. The 'Mvar Capability Curve' table is empty. The 'Wind Control Mode' is set to 'None' and the 'Power Factor' is 1.0000. The 'Regulated Bus Number' is 1, 'SetPoint Voltage' is 1.000000, 'SetPoint Voltage Tol' is 0.000000, and 'Remote RegFactor' is 100.0. The 'Line Drop Compensation' section shows 'Use LDC' as 'No', 'Xcomp' as 0.000100, and 'Rcomp' as 0.000000. The 'Voltage Droop Control' section is empty.

	MW	Min Mvar	Max Mvar
1			
2			
3			
4			
5			

Figure:- Transformer connected between (Bus 1-Bus 5)

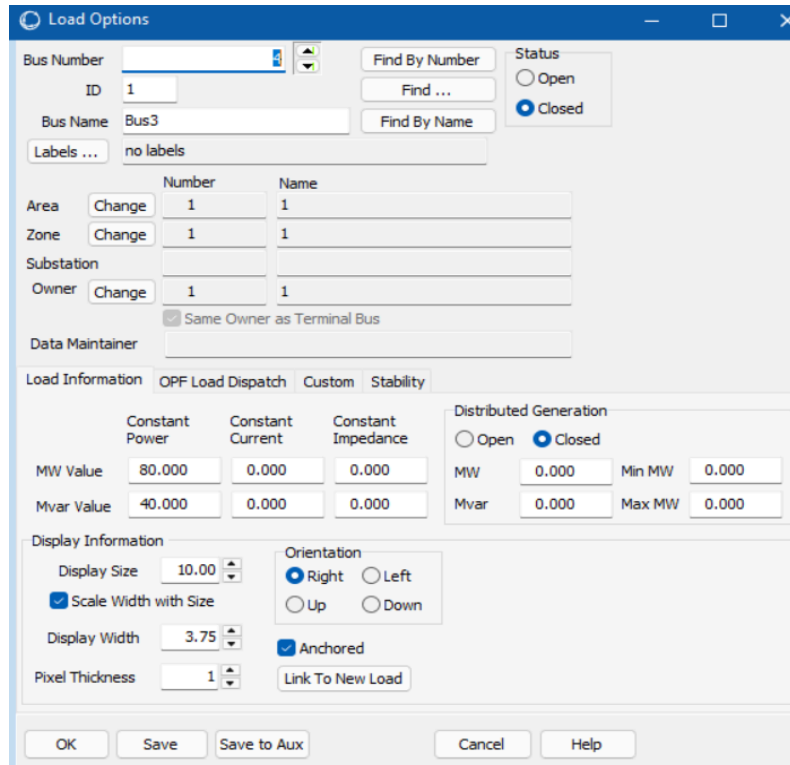


The 'Generator Options' dialog box is shown for Bus 3. The 'Status' is set to 'Closed'. The 'Generator MVA Base' is 100.00. The 'Fuel Type' is 'UN (Unknown)' and the 'Unit Type' is empty. The 'Display Information' tab is selected. The 'Power Control' section shows 'MW Setpoint' at 520.000, 'MW Output' at 520.000, and 'Part. Factor' at 10.00. The 'Voltage Control' section shows 'Mvar Output' at 252.979, 'Min Mvars' at -240.000, and 'Max Mvars' at 400.000. The 'Mvar Capability Curve' table is empty. The 'Wind Control Mode' is set to 'None' and the 'Power Factor' is 1.0000. The 'Regulated Bus Number' is 4, 'SetPoint Voltage' is 1.050000, 'SetPoint Voltage Tol' is 0.000000, and 'Remote RegFactor' is 100.0. The 'Line Drop Compensation' section shows 'Use LDC' as 'No', 'Xcomp' as 0.000100, and 'Rcomp' as 0.000000. The 'Voltage Droop Control' section is empty.

	MW	Min Mvar	Max Mvar
1			
2			
3			
4			
5			

Figure:- Transformer connected between (Bus 4-Bus 3)

Load Data setup



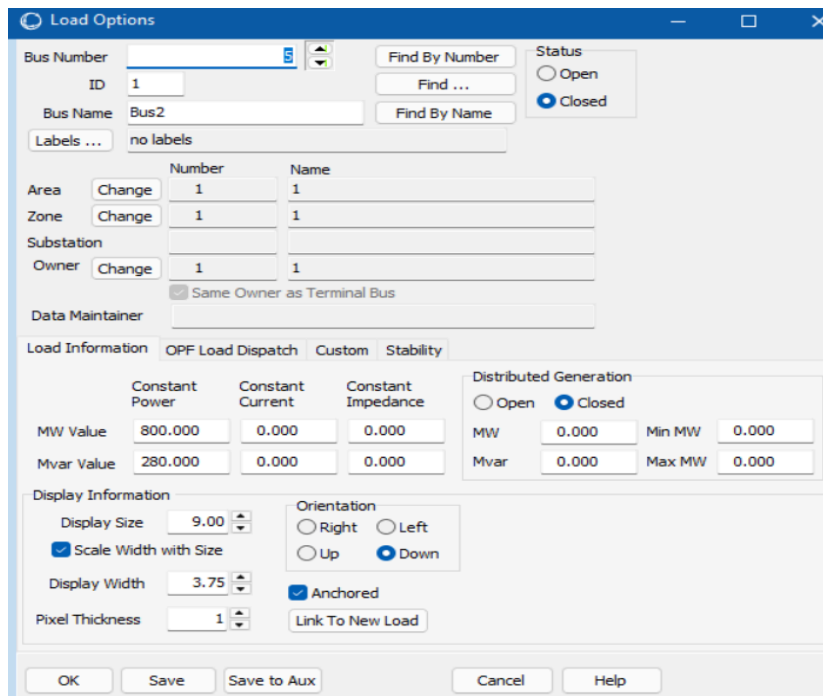
The 'Load Options' dialog box is shown for BUS 3. The 'Bus Number' is 1, 'Bus Name' is Bus3, and 'Status' is Closed. The 'Area' is 1, 'Zone' is 1, 'Substation' is empty, and 'Owner' is 1. The 'Data Maintainer' is empty. The 'Load Information' tab is selected, showing 'Constant Power' with MW Value 80.000 and Mvar Value 40.000, 'Constant Current' with 0.000, and 'Constant Impedance' with 0.000. The 'Distributed Generation' is set to Closed. The 'Display Information' section shows 'Display Size' 10.00, 'Scale Width with Size' checked, 'Display Width' 3.75, 'Pixel Thickness' 1, 'Orientation' set to Right, and 'Anchored' checked. The 'Link To New Load' button is available.

Area	Number	Name
1	1	1

Constant Power	Constant Current	Constant Impedance
MW Value: 80.000	0.000	0.000
Mvar Value: 40.000	0.000	0.000

Distributed Generation	Min MW	Max MW
0.000	0.000	0.000

Figure:- Load besides BUS 3



The 'Load Options' dialog box is shown for BUS 2. The 'Bus Number' is 1, 'Bus Name' is Bus2, and 'Status' is Closed. The 'Area' is 1, 'Zone' is 1, 'Substation' is empty, and 'Owner' is 1. The 'Data Maintainer' is empty. The 'Load Information' tab is selected, showing 'Constant Power' with MW Value 800.000 and Mvar Value 280.000, 'Constant Current' with 0.000, and 'Constant Impedance' with 0.000. The 'Distributed Generation' is set to Closed. The 'Display Information' section shows 'Display Size' 9.00, 'Scale Width with Size' checked, 'Display Width' 3.75, 'Pixel Thickness' 1, 'Orientation' set to Down, and 'Anchored' checked. The 'Link To New Load' button is available.

Area	Number	Name
1	1	1

Constant Power	Constant Current	Constant Impedance
MW Value: 800.000	0.000	0.000
Mvar Value: 280.000	0.000	0.000

Distributed Generation	Min MW	Max MW
0.000	0.000	0.000

Figure:- Load besides BUS 2

Transmission line Data setup

The Branch Options dialog box is shown for a transmission line. The 'Line' tab is active. The 'From Bus' is 2 and the 'To Bus' is 3. The 'Circuit' is 1. The 'Name' is Bus5 and the 'Area Name' is 1 (1). The 'Nominal kV' is 345.0. The 'Status' is Open. The 'Branch Device Type' is Line. The 'Length' is 0.00. The 'Normal Status' is Closed. The 'Per Unit Impedance Parameters' are: Series Resistance (R) 0.002250, Series Reactance (X) 0.025000, Shunt Charging (B) 0.440000, and Shunt Conductance (G) 0.000000. The 'MVA Limits' are: Limit A 1200.000, Limit B 0.000, Limit C 0.000, Limit D 0.000, Limit E 0.000, Limit F 0.000, Limit G 0.000, Limit H 0.000, Limit I 0.000, Limit J 0.000, and Limit K 0.000. The 'Default Owner (Same as From Bus)' checkbox is checked. The 'Labels ...' field is set to 'no labels'. The 'Convert Line to Transformer' and 'Exchange From and To Buses' buttons are visible. The 'D-FACTS Devices on the Line' checkbox is unchecked. The 'Has D-FACTS' checkbox is also unchecked.

Line	From Bus	To Bus	Circuit
Number	2	3	1
Name	Bus5	Bus-4	
Area Name	1 (1)	1 (1)	
Nominal kV	345.0	345.0	

Per Unit Impedance Parameters	MVA Limits
Series Resistance (R)	Limit A
Series Reactance (X)	Limit B
Shunt Charging (B)	Limit C
Shunt Conductance (G)	Limit D
	Limit E
	Limit F
	Limit G
	Limit H
	Limit I
	Limit J
	Limit K

Figure:- Transmission line between BUS 5-BUS 4

The Branch Options dialog box is shown for a transmission line. The 'Line' tab is active. The 'From Bus' is 2 and the 'To Bus' is 5. The 'Circuit' is 1. The 'Name' is Bus5 and the 'Area Name' is 1 (1). The 'Nominal kV' is 345.0. The 'Status' is Open. The 'Branch Device Type' is Line. The 'Length' is 0.00. The 'Normal Status' is Closed. The 'Per Unit Impedance Parameters' are: Series Resistance (R) 0.004500, Series Reactance (X) 0.050000, Shunt Charging (B) 0.880000, and Shunt Conductance (G) 0.000000. The 'MVA Limits' are: Limit A 1200.000, Limit B 0.000, Limit C 0.000, Limit D 0.000, Limit E 0.000, Limit F 0.000, Limit G 0.000, Limit H 0.000, Limit I 0.000, Limit J 0.000, and Limit K 0.000. The 'Default Owner (Same as From Bus)' checkbox is checked. The 'Labels ...' field is set to 'no labels'. The 'Convert Line to Transformer' and 'Exchange From and To Buses' buttons are visible. The 'D-FACTS Devices on the Line' checkbox is unchecked. The 'Has D-FACTS' checkbox is also unchecked.

Line	From Bus	To Bus	Circuit
Number	2	5	1
Name	Bus5	Bus2	
Area Name	1 (1)	1 (1)	
Nominal kV	345.0	345.0	

Per Unit Impedance Parameters	MVA Limits
Series Resistance (R)	Limit A
Series Reactance (X)	Limit B
Shunt Charging (B)	Limit C
Shunt Conductance (G)	Limit D
	Limit E
	Limit F
	Limit G
	Limit H
	Limit I
	Limit J
	Limit K

Figure:- Transmission line between BUS 5-BUS 2

Branch Options

Line	From Bus	To Bus	Circuit
Number	3	5	1
Name	Bus-4	Bus2	
Area Name	1 (1)	1 (1)	
Nominal kV	345.0	345.0	

Labels ... no labels

Find By Numbers Find By Names Find ...

☒ From End Metered

☒ Default Owner (Same as From Bus)

Display Parameters Fault Info Owner, Area, Zone, Sub Custom Stability Geography

Status

☐ Open

☒ Closed

Branch Device Type

Line

☐ Allow Consolidation

Length 0.00

Calculate Impedances >

Normal Status

☐ Open

☒ Closed

Per Unit Impedance Parameters

Series Resistance (R)	0.009000
Series Reactance (X)	0.100000
Shunt Charging (B)	1.720000
Shunt Conductance (G)	0.000000

☐ Has Line Shunts

Line Shunts

MVA Limits

Limit A	1200.000
Limit B	0.000
Limit C	0.000
Limit D	0.000
Limit E	0.000
Limit F	0.000
Limit G	0.000
Limit H	0.000
Limit I	0.000
Limit J	0.000
Limit K	0.000

Convert Line to Transformer

Exchange From and To Buses

D-FACTS Devices on the Line

☐ Has D-FACTS

OK Save Save to Aux Cancel Help

Figure:- Transmission line between BUS 2-BUS 4

The diagram illustrates the IEEE 30 bus test system with the following data:

- Bus1:** 394 MW, -82 Mvar, 402 MVA. Slack bus.
- Bus2:** 800 MW, 280 Mvar. 0.84 pu, -22.09 Deg, 290.86 kV.
- Bus3:** 520 MW, 253 Mvar. 1.05 pu, -0.61 Deg, 15.75 kV.
- Bus4:** 134 MW, 165 Mvar, 212 MVA. 1.03 pu, -2.86 Deg, 354.44 kV.
- Bus5:** 133 MW, -196 Mvar, 237 MVA. 0.98 pu, -4.54 Deg, 337.80 kV.

Transmission line data and losses:

- Line 1-2:** 2 MW, -139 Mvar. Losses: 17 MW, 116 Mvar.
- Line 1-5:** 391 MW, -58 Mvar, 396 MVA. Losses: 2 MW, -139 Mvar.
- Line 2-3:** 508 MW, -138 Mvar, 526 MVA. Losses: 12 MW, -22 Mvar.
- Line 2-4:** 304 MW, 120 Mvar, 327 MVA. Losses: 12 MW, -22 Mvar.
- Line 3-4:** 438 MW, -284 Mvar, 522 MVA. Losses: 2 MW, -71 Mvar.
- Line 4-5:** 133 MW, -196 Mvar, 237 MVA. Losses: 1 MW, -32 Mvar.

Losses at various points:

- Losses: 2 MW, -139 Mvar.
- Losses: 17 MW, 116 Mvar.
- Losses: 12 MW, -22 Mvar.
- Losses: 2 MW, -71 Mvar.
- Losses: 1 MW, -32 Mvar.

b. Use PowerWorld to determine the Ybus of this system.

Answer:-

For Ybus matrices ,we have to do a Fault analysis in the system .

Set fault values,

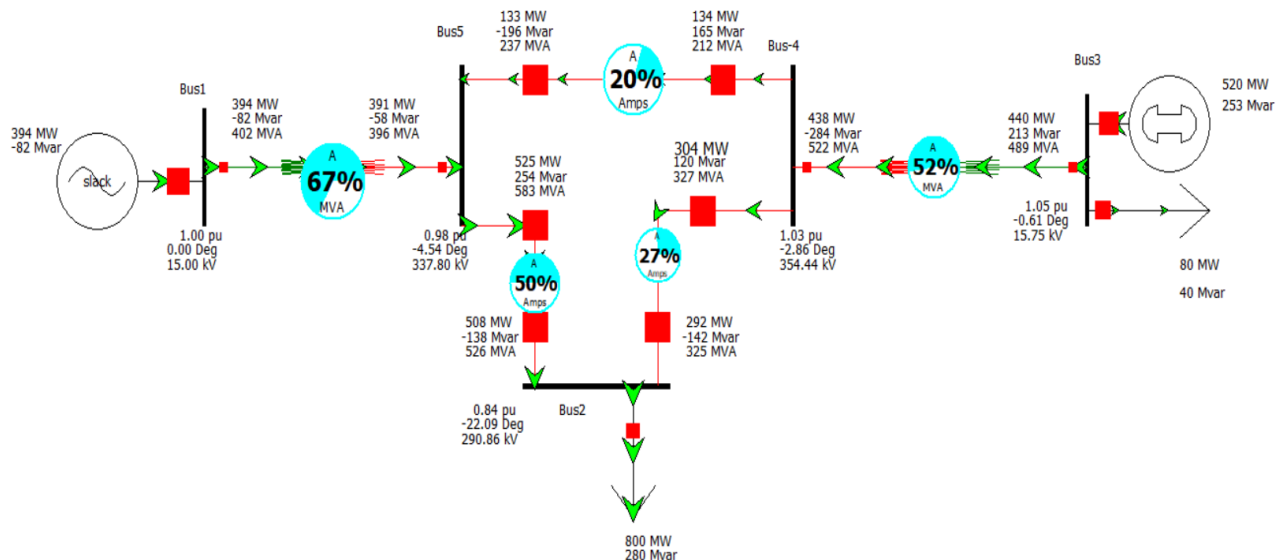
$R = 0.01$

$X = 0.02$

Here is the Ybus matrices,

	Number	Name	Bus 1	Bus 2	Bus 3	Bus 4	Bus 5
1	1	Bus1	3.73 - j49.00	-3.73 + j49.72			
2	2	Bus5	-3.73 + j49.72	9.09 - j108.58	-3.57 + j39.68		-1.79 + j19.84
3	3	Bus4		-3.57 + j39.68	11.92 - j147.08	-7.46 + j99.44	-0.89 + j9.92
4	4	Bus3			-7.46 + j99.44	8.18 - j100.80	
5	5	Bus2		-1.79 + j19.84	-0.89 + j9.92		13.93 - j32.40

Name	Bus 1	Bus 2	Bus 3	Bus 4	Bus 5
Bus 1	3.73-j49.00	-3.73+j49.72			
Bus 5	-3.73+j49.72	9.09-j108.58	-3.57+j39.68		-1.79+j19.84
Bus 4		-3.57+j39.68	11.92-j147.08	-7.46+j99.44	-0.89+j9.92
Bus 3			-7.46+j99.44	8.18-j100.80	
Bus 2		-1.79+j19.84	-0.89+j9.92		13.93-j32.40



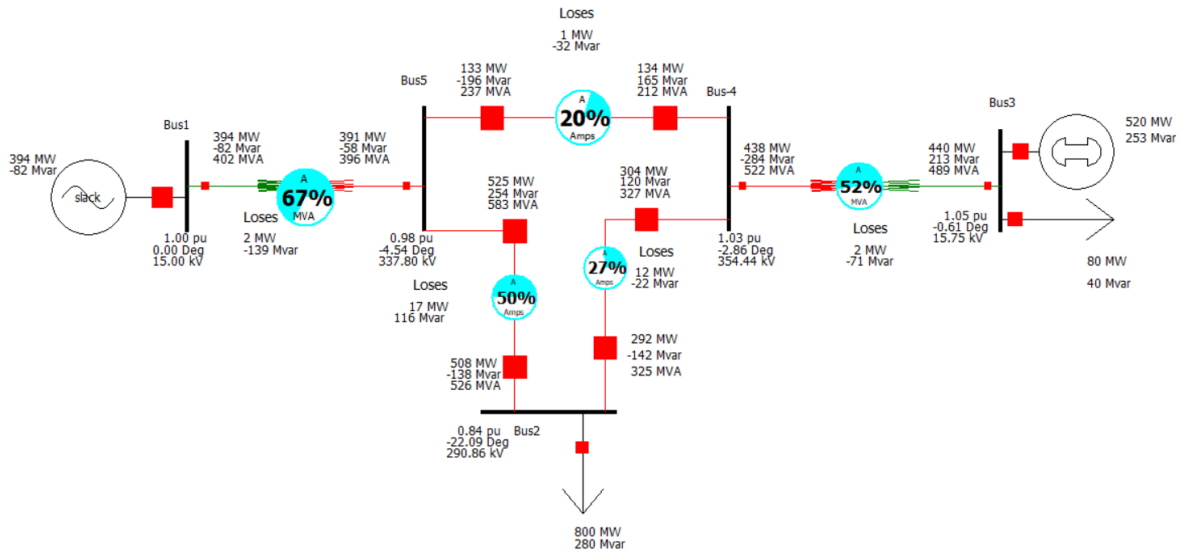
State Variables

	Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)
1	1	Bus1	1	15.00	1.00000	15.000	0.00
2	2	Bus5	1	345.00	0.97912	337.796	-4.54
3	3	Bus-4	1	345.00	1.02737	354.442	-2.86
4	4	Bus3	1	15.00	1.05000	15.750	-0.61
5	5	Bus2	1	345.00	0.84307	290.858	-22.09

Bus	Voltage(pu)	Angle(degree)	Voltage(kV)
1(slack)	1	0	15
2(Load)	0.84	-22.09	290.86
3(Load)	1.05	-0.61	15.75
4(PV)	1.03	-2.86	354.44
5	0.98	-4.54	337.80

Calculate Losses:

Line Losses



Bus to Bus	ΔMW	$\Delta MVAR$	ΔMVA
1-5	2	-139	6
4-5	1	-32	-25
5-2	17	116	57
4-2	12	-22	2
3-4	2	-71	-33
	Total =34	Total = -148	Total = 6

Power Consumed by Loads

- Load at Bus 2: 800 MW and 280 Mvar
- Load at Bus 3: 80 MW and 40 Mvar

Transformer Losses

- Transformer T1 (1-5): 2 MW loss, 6 MVA, -139 Mvar
- Transformer T2 (3-4): 2 MW loss, -71 Mvar, -33 MVA

